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Assignment 9: RAID implementation and performance analysis

<https://github.com/elovric13/eutech/tree/RAID-implementation-and-performance-analysis-elovric13>

1. Research and explain the differences between RAID 0, RAID 1, RAID 5, and RAID 10

RAID is a technology that combines multiple physical disks into one logical unit to improve performance, redundancy, or both. Different RAID levels offer different trade-offs between speed, fault tolerance, and storage efficiency.

RAID Level	Min. Disks	Redundancy	Performance	Storage efficiency
RAID 0 (Striping)	2	None — 1 disk failure = total data loss	Best read and write speed	100%
RAID 1 (Mirroring)	2	1 disk failure tolerated	Good read, normal write	50%
RAID 5 (Striping + Parity)	3	1 disk failure tolerated	Good read, slightly slower write	67–94%
RAID 10 (Mirror + Stripe)	4	Multiple failures tolerated	Best combined performance	50%

RAID 0 offers the best performance but no fault tolerance, making it unsuitable for critical data. RAID 1 provides full redundancy by mirroring data across two disks but uses only half the total capacity. RAID 5 balances performance, redundancy and storage efficiency by distributing parity information across all disks. RAID 10 combines mirroring and striping for the best reliability and speed but requires at least four disks and uses only 50% of total capacity.

2. Create a VM with at least three additional virtual disks and configure two RAID levels

An Ubuntu virtual machine was used with three additional virtual disks of 5GB each: sdb, sdc, and sdd. The mdadm tool was used to configure the RAID arrays. Two RAID levels were implemented: RAID 1 using sdb and sdc, and RAID 5 using sdb, sdc, and sdd.

```
ubuntu26@ubuntu26:~$ lsblk -o NAME,SIZE,FSTYPE,TYPE,MOUNTPOINT
NAME        SIZE FSTYPE  TYPE MOUNTPOINT
loop0       66.8M squashfs loop  /snap/core24/1587
loop1        4K squashfs loop  /snap/bare/5
loop2       48.4M squashfs loop  /snap/snapd/26382
loop3       66.8M squashfs loop  /snap/core24/1643
loop4       395M squashfs loop  /snap/mesa-2404/1165
loop5       273.7M squashfs loop  /snap/firefox/8107
loop6       606.1M squashfs loop  /snap/gnome-46-2404/153
loop7       91.7M squashfs loop  /snap/gtk-common-themes/1535
loop8       49.3M squashfs loop  /snap/snapd/26865
sda          20G                disk
├─sda1         1M                part
├─sda2         1.8G ext4          part  /boot
├─sda3       18.2G LVM2_member part
└─ubuntu--vg-ubuntu--lv 10G ext4          lvm   /
sdb           5G                disk
sdc           5G                disk
sdd           5G                disk
sr0          1024M                rom
```

Figure 1: lsblk output showing sdb, sdc, sdd disks attached to the VM

- **RAID 1 – md0 (sdb + sdc)**

RAID 1 was created using the following command:

```
sudo mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdb /dev/sdc
```

The array was formatted with the ext4 filesystem and mounted at /raid1.

```
ubuntu26@ubuntu26:~$ sudo mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdb /dev/sdc
mdadm: Note: this array has metadata at the start and
      may not be suitable as a boot device.  If you plan to
      store '/boot' on this device please ensure that
      your boot-loader understands md/v1.x metadata, or use
      --metadata=0.90
mdadm: size set to 5237760K
Continue creating array? yes
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
```

Figure 2: RAID 1 array creation

```
ubuntu26@ubuntu26:~$ cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md0 : active raid1 sdc[1] sdb[0]
      5237760 blocks super 1.2 [2/2] [UU]
      [=====>....] resync = 80.2% (4201856/5237760) finish=0.0min speed=210092K/sec

unused devices: <none>
```

Figure 3: /proc/mdstat showing md0 active raid1 with [UU] status

```
ubuntu26@ubuntu26:~$ sudo mkfs.ext4 /dev/md0
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 1309440 4k blocks and 327680 inodes
Filesystem UUID: 21b028ce-deca-4aa1-8db5-bf14463dda15
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

Figure 4: mkfs.ext4 /dev/md0 formatting

```

ubuntu26@ubuntu26:~$ sudo mkdir /raid1
ubuntu26@ubuntu26:~$ sudo mount /dev/md0 /raid1
ubuntu26@ubuntu26:~$ df -h

```

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	387M	4.9M	382M	2%	/run
/dev/mapper/ubuntu--vg-ubuntu--lv	9.8G	7.7G	1.6G	84%	/
tmpfs	1.9G	0	1.9G	0%	/dev/shm
tmpfs	5.0M	8.0K	5.0M	1%	/run/lock
/dev/sda2	1.8G	213M	1.4G	13%	/boot
tmpfs	387M	80K	387M	1%	/run/user/1000
tmpfs	387M	92K	387M	1%	/run/user/123
/dev/md0	4.9G	24K	4.6G	1%	/raid1

Figure 5: df -h showing /dev/md0 mounted at /raid1

- **RAID 5 – md1 (sdb + sdc + sdd)**

RAID 5 was created using the following command:

```

sudo mdadm --create --verbose /dev/md1 --level=5 --raid-devices=3 /dev/sdb
/dev/sdc /dev/sdd

```

The array was formatted with ext4 and mounted at /raid5.

```

ubuntu26@ubuntu26:~$ sudo mdadm --create --verbose /dev/md1 \
--level=5 \
--raid-devices=3 \
/dev/sdb /dev/sdc /dev/sdd
mdadm: layout defaults to left-symmetric
mdadm: layout defaults to left-symmetric
mdadm: chunk size defaults to 512K
mdadm: size set to 5237760K
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md1 started.

```

Figure 6: RAID 5 array creation

```

ubuntu26@ubuntu26:~$ cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md1 : active raid5 sdd[3] sdc[1] sdb[0]
      10475520 blocks super 1.2 level 5, 512k chunk, algorithm 2 [3/3] [UUU]

unused devices: <none>

```

Figure 7: /proc/mdstat showing md1 active raid5 with [UUU] status

```
ubuntu26@ubuntu26:~$ df -h
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                     387M        4.9M   382M    2% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 9.8G       7.7G    1.6G   84% /
tmpfs                     1.9G         0    1.9G    0% /dev/shm
tmpfs                     5.0M        8.0K    5.0M    1% /run/lock
/dev/sda2                 1.8G       213M    1.4G   13% /boot
tmpfs                     387M        80K    387M    1% /run/user/1000
tmpfs                     387M        92K    387M    1% /run/user/123
/dev/md1                  9.8G       2.1G    7.3G   22% /raid5
```

Figure 8: df -h showing /dev/md1 mounted at /raid5

3. Install an operating system on a RAID volume and validate its functionality

Functionality was validated through successful mounting of both arrays confirmed in Figures 5 and 8, and through the fio performance tests shown in Figures 10 and 11.

4. Run performance tests using fio

Performance tests were run on both RAID arrays using the fio tool with a 60-second mixed read/write workload to measure throughput on each array.

```
Run status group 0 (all jobs):
  READ: bw=108MiB/s (113MB/s), 49.6MiB/s-60.2MiB/s (52.0MB/s-63.1MB/s), io=6684MiB (7009MB), run=600
01-61924msec
  WRITE: bw=108MiB/s (113MB/s), 49.5MiB/s-60.3MiB/s (51.9MB/s-63.2MB/s), io=6682MiB (7007MB), run=600
01-61924msec

Disk stats (read/write):
  md0: ios=11176/5807, sectors=5530704/13409544, merge=0/0, ticks=22584/566747, in_queue=589331, uti
l=99.76%, aggrrios=5572/5785, aggrsectors=2765368/13409570, aggrmerge=16/27, aggrticks=11185/481794, a
ggrin_queue=492979, aggrutil=99.42%
  sdb: ios=7344/5785, sectors=3596544/13409570, merge=22/27, ticks=13274/480401, in_queue=493674, uti
l=99.42%
  sdc: ios=3801/5785, sectors=1934192/13409570, merge=10/27, ticks=9096/483188, in_queue=492284, uti
l=99.28%
```

Figure 9: fio output for RAID 1

```
Disk stats (read/write):
  md1: ios=26622/13645, sectors=18083600/26995816, merge=0/0, ticks=98970/923835, in_queue=1022805,
util=97.92%, aggrrios=7575/16291, aggrsectors=6088245/13563334, aggrmerge=5135/1679182, aggrticks=2863
7/115646, aggrin_queue=144284, aggrutil=79.54%
  sdd: ios=7591/16288, sectors=6088656/13561110, merge=5229/1678856, ticks=28466/119408, in_queue=147
873, util=78.41%
  sdb: ios=7572/16283, sectors=6087904/13562014, merge=5358/1679002, ticks=28762/112950, in_queue=141
713, util=79.54%
  sdc: ios=7564/16302, sectors=6088176/13566878, merge=4818/1679690, ticks=28685/114580, in_queue=143
266, util=77.11%
```

Figure 10: fio output for RAID 5

5. Compare the redundancy, performance, and reliability of each RAID level

Based on the configuration and test results, the two RAID levels were compared.

	RAID 1	RAID 5
Disks used	2 (sdb, sdc)	3 (sdb, sdc, sdd)
Usable capacity	4.9 GB (50% of 10 GB)	9.8 GB (67% of 15 GB)
Redundancy	1 disk failure tolerated	1 disk failure tolerated
Read performance	108 MiB/s	Lower due to parity overhead
Write performance	108 MiB/s	Slower due to parity calculation
Disk utilization	~99%	~78–79%
Failure simulation	Array stayed active with [_U] status	Array stayed active with [_UU] status

RAID 1 showed higher disk utilization and throughput in the fio test because all I/O is split between only two mirrored disks. RAID 5 showed lower utilization per disk because the workload is spread across three disks with additional parity calculations. However RAID 5 provided nearly double the usable storage capacity compared to RAID 1 with the same number of disks.

Both arrays successfully survived a simulated disk failure. In RAID 1, marking sdb as faulty left the array running on sdc alone with a degraded [_U] status. In RAID 5, marking sdb as faulty left the array running in degraded mode with [_UU] status, continuing to serve data using the parity information stored on the remaining disks.

```
ubuntu26@ubuntu26:~$ sudo mdadm /dev/md0 --fail /dev/sdb
mdadm: set /dev/sdb faulty in /dev/md0
ubuntu26@ubuntu26:~$
ubuntu26@ubuntu26:~$
ubuntu26@ubuntu26:~$ cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md0 : active raid1 sdc[1] sdb[0](F)
      5237760 blocks super 1.2 [2/1] [_U]

unused devices: <none>
ubuntu26@ubuntu26:~$
```

Figure 11: RAID 1 degraded mode after sdb failure

```
ubuntu26@ubuntu26:~$ sudo mdadm /dev/md1 --fail /dev/sdb
mdadm: set /dev/sdb faulty in /dev/md1
ubuntu26@ubuntu26:~$
ubuntu26@ubuntu26:~$ cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md1 : active raid5 sdd[3] sdc[1] sdb[0](F)
      10475520 blocks super 1.2 level 5, 512k chunk, algorithm 2 [3/2] [_UU]

unused devices: <none>
```

Figure 12: RAID 5 degraded mode after sdb failure